

PRST Class (6/10/2020)

Markov Inequality

X is a non-negative rv

$$P(X \geq a) \leq \frac{E(X)}{a} \quad \text{where } a > 0.$$

Chebyshev Inequality

If X is a rv with $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$, ($\sigma^2 > 0$)

Then

$$P(|X - \mu| \geq b) \leq \frac{\text{Var}(X)}{b^2}, \quad b > 0$$

Ex 1. Suppose the average grade in a quiz is 70%. Give an upper bound on the proportions of students who scored at least 90%.

Sol: X - grade in the quiz

$$E(X) = 0.7$$

$$P(X \geq 0.90) \leq \frac{E(X)}{0.9} = \frac{0.7}{0.9} = \frac{7}{9}$$

Ex 2. Suppose a coin is flipped 20 times.

$$P(H) = 0.5$$

1.1.1: Let that

Ex 2.

$$P(H) = 0.2$$

Find a bound for the probability that head will come at least 16 times.

Sol: — X : number heads appearing out of 20 tosses.

$$P(X \geq 16)$$

$$P(X = x)$$

$$\begin{matrix} x=0 \\ x=1 \\ \vdots \\ x=20 \end{matrix}$$

$$P(X=0) = (0.8)^{20}$$

$$P(X=1) =$$

$$P(X=0) = P(T)P(T)\dots P(T)$$

$$= P(T \cap T \cap T \dots \cap T)$$

$$P(X=1) = \binom{20}{1} (0.2)^1 (0.8)^{20-1}$$

$$\frac{T}{1} \frac{T}{2} \frac{T}{3} \dots \frac{T}{20}$$

$$\begin{cases} P(H) = 0.2 \\ P(T) = 1 - 0.2 \\ = 0.8 \end{cases}$$

$$\begin{matrix} TT \\ TH \\ HT \\ HH \end{matrix} \quad \begin{aligned} P(TT) &= P(T)P(T) \\ &= (0.8)(0.8) \end{aligned}$$

$$\begin{matrix} H & T & T \\ T & T & H \\ H & T & T \end{matrix} \quad \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$${}^n C_r = \binom{n}{r}$$

$$\binom{3}{1} = 3$$

$$C_r = \binom{n}{r}$$

$$P(T, H, T) = P(T)P(H)P(T)$$

$$P(\text{getting only 1 H}) = 3 \times P(T)^2 P(H)$$

$$= \binom{3}{1} \times P(T)^2 P(H)$$

$$P(X=2) = \binom{20}{2} P(H)^2 P(T)^{18}$$

$$= \binom{20}{2} (0.2)^2 (0.8)^{18}$$

$$P(X=k) = \binom{20}{k} P(H)^k P(T)^{20-k}$$

$$= \binom{20}{k} (0.2)^k (0.8)^{20-k}$$

$$E(X) = \sum_{k=0}^{20} k \cdot P(X=k) = 20 \times (0.2) = 4$$

$$P(X \geq 16) \leq \frac{E(X)}{16} = \frac{4}{16} = \frac{1}{4}$$

20 (n).

X : denotes the number of heads coming out of n tosses of a coin at.

$$P(H) = p$$

$$0 < p < 1$$

$$n = 0, 1, \dots, n$$

$$P(H) = p$$

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x} \quad x=0,1,\dots,n.$$

$$E(X) = n \cdot p$$

$$V(X) = n \cdot p(1-p).$$

Eg 3.

Suppose a fair coin is flipped 100 times.
Find a bound on the prob that the number
of times the coin lands on heads
is at least 60 or at most 40.

Sol

$$P(X=x) = \binom{100}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{100-x}$$

$$x=0, \dots, 100$$

$$E(X) = 50$$

$$V(X) = 25$$

$$P(X \geq 60 \cup X \leq 40)$$

$$= P(X - E(X) \geq 60 - E(X) \cup X - E(X) \leq 40 - E(X))$$

$$= P(\underbrace{X - 50}_{E(X)} \geq 10 \cup \underbrace{X - 50}_{E(X)} \leq -10)$$

$$= P(|X - 50| \geq 10) \quad (=6)$$

$$\leq \frac{V(X)}{10^2}$$

$$= \frac{25}{100} = \frac{1}{4}.$$

$$\begin{cases} x \geq 1 \text{ or} \\ x \leq -1 \\ |x| \geq 1 \end{cases}$$

$$P(X \geq 60 \cup X \leq 30)$$

$$= P(X \geq 60 \cup X \leq 20)$$

$$\overline{P(X \geq 60 \cup X \leq 30)} \\ = P(X - 50 \geq 10 \cup X - 50 \leq -20)$$

Eg 4

X is rv $E(X) = 5$, $V(X) = 3$.

$$P(|X - 5| < h) \geq 0.99$$

$h = ?$

$$A = \{|X - 5| < h\}$$

$$A^c = \{|X - 5| \geq h\}$$

$$P(A) = 1 - P(A^c).$$